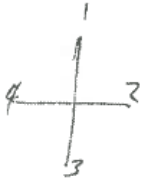
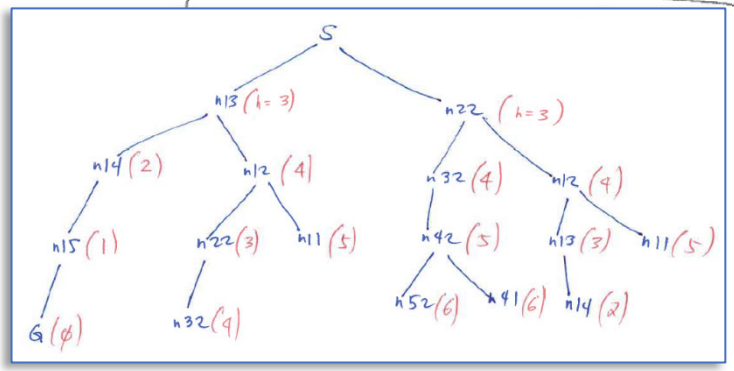
# MAZE PROBLEM

A* with Strict Expanded list (4-connected Gridworld)

h = Manhattan distance




| Step | Deq | Enq | Expanded |
|---|---|---|---|
| 0 | | ~~(25)~~ | |
| 1 | (2 5) | ~~(4 n12, 5)~~ (4 n22, 5) | 5 |
| 2 | (4 n13, 5) | ~~(4 n14, n15, 5)~~ ~~(6 n12, n13, 5)~~ (4 n22, 5) | 5, n13 |
| 3 | (4 n14, n13, 5) | ~~(4 n15, n14, n13, 5)~~ (6 n12, n13, 5) (4, n22, 5) | 5, n13, n14, |
| 4 | (4 n15, n14, n13, 5) | (4G, n15, n14, n13, 5) (6 n12, n13, 5) (4, n22, 5) | ", n15 |
| 5 | (4G, n15, n14, n13, 5) | path to Goal is found! | |
| 6 | | | |




| | | | | |
|---|---|---|---|---|
| n51 | n52 | n53 | n54 | n55 |
| n41 | n42 | | n44 | n45 |
| | n32 | | n34 | n35 |
| | n22 | S | | G |
| n11 | n12 | n13 | n14 | n15 |

N W E S

Successor generator = {N, E, S, W}